

# Three-body-ODE

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## 1 Introduction

The basic mathematics of circular motion. Consider a circle with radius  $r$ , circumference  $L$ , arc length  $S$ , and central angle  $\theta$ , where  $\theta$  is measured in radians.

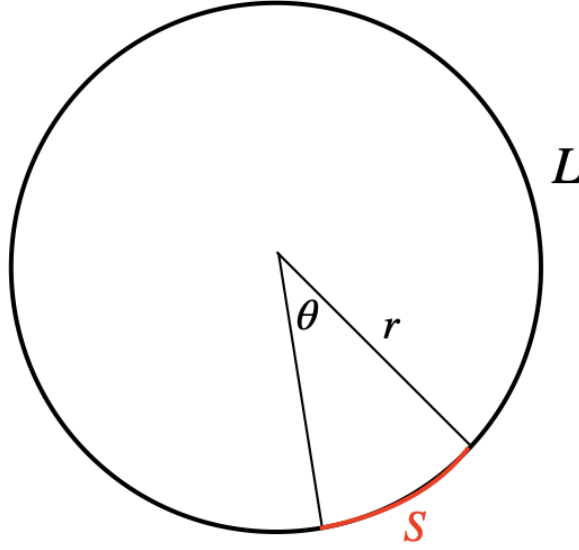


Figure 1: Relationship between the central angle and arc length in a circle.

### 1.1 Radian and Arc Length

The complete circumference is  $L = 2\pi r$ . Now suppose the central angle is  $\theta$ , with corresponding arc length is  $S$ . Since  $\frac{\theta}{2\pi} = \frac{S}{L}$ , we have  $S = \frac{\theta}{2\pi}L$ . Substitute  $L = 2\pi r$ , we obtain  $S = \frac{\theta}{2\pi}2\pi r = r\theta$ .

### 1.2 Angular Velocity

For circular motion with a constant radius  $r$ , suppose the object moves through a small angle  $\Delta\theta$ . The corresponding change in arc length is  $\Delta S = r\Delta\theta$ . The tangential speed is defined as  $v = \lim_{t \rightarrow 0} \frac{\Delta S}{\Delta t} = \lim_{t \rightarrow 0} \frac{r\Delta\theta}{\Delta t}$ . Since the angular velocity is defined as  $\omega = \frac{\Delta\theta}{\Delta t}$ , we obtain  $v = r\omega$ .

### 1.3 Polar Unit Vectors

In two-dimensional polar coordinates, the position vector can be expressed as  $\vec{r} = r\hat{r}$ , where  $r$  is the distance from the origin and  $\hat{r}$  is the radial unit vector. The tangential unit vector  $\hat{\theta}$ , which is perpendicular to  $\hat{r}$  and points in the direction of increasing  $\theta$ .

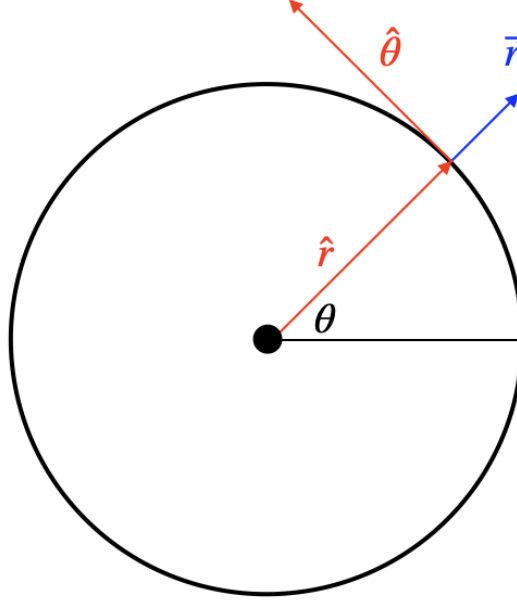


Figure 2: Radial and tangential unit vectors in polar coordinates

In Cartesian coordinates, the polar unit vectors can be expressed in terms of the Cartesian unit vectors  $\hat{i}$  and  $\hat{j}$  as  $\hat{r} = \cos \theta \hat{i} + \sin \theta \hat{j}$  and  $\hat{\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j}$ . Differentiate  $\hat{r}$  with respect to  $\theta$ :

$$\begin{aligned}\frac{d\hat{r}}{d\theta} &= \frac{d(\cos \theta \hat{i} + \sin \theta \hat{j})}{d\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j} = \hat{\theta}, \\ \frac{d\hat{\theta}}{d\theta} &= \frac{d(-\sin \theta \hat{i} + \cos \theta \hat{j})}{d\theta} = -\cos \theta \hat{i} - \sin \theta \hat{j} = -\hat{r}.\end{aligned}$$

Therefore, the derivatives of the polar unit vectors with respect to  $\theta$  are

$$\frac{d\hat{r}}{d\theta} = \hat{\theta} \quad \text{and} \quad \frac{d\hat{\theta}}{d\theta} = -\hat{r}.$$

#### 1.4 Velocity in Polar Coordinates

The position vector is  $\vec{r} = r\hat{r}$ . Velocity is the time derivative of position  $\vec{v} = \frac{d\vec{r}}{dt}$ . By the product rule, we have  $\vec{v} = \frac{d}{dt}(r\hat{r}) = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$ . Then by the chain rule,  $\frac{d\hat{r}}{dt} = \frac{d\hat{r}}{d\theta}\frac{d\theta}{dt} = \hat{\theta}\dot{\theta}$ . Therefore,  $\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$ .

The first component,  $v_r = \dot{r}$ , is the radial velocity, which describes how quickly the object moves toward or away from the origin. The second component,  $r\dot{\theta}$ , is the tangential (circumferential) velocity, which describes how quickly the object moves around the origin.

For a circular orbit,  $r$  is constant, which means  $\dot{r} = 0$ . Therefore,  $\vec{v} = r\dot{\theta}\hat{\theta}$ . Since  $\omega = \dot{\theta}$ , we recover  $v = r\omega$ .

#### 1.5 Acceleration in Polar Coordinates

Acceleration is the time derivative of velocity  $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(\dot{r}\hat{r} + r\dot{\theta}\hat{\theta})$ . In the first term,  $\frac{d}{dt}(\dot{r}\hat{r}) = \ddot{r}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} = \ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta}$ . In the second term,  $\frac{d}{dt}(r\dot{\theta}\hat{\theta}) = \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt}$ . Then by the chain rule,  $\frac{d\hat{\theta}}{dt} = \frac{d\hat{\theta}}{d\theta}\frac{d\theta}{dt} = -\hat{r}\dot{\theta}$ . Therefore,  $\vec{a} = \ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} - r\dot{\theta}^2\hat{r} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$ .

Therefore, the acceleration can be decomposed into the radial and tangential components:

$$a_r = (\ddot{r} - r\dot{\theta}^2), \quad a_\theta = (2\dot{r}\dot{\theta} + r\ddot{\theta})$$

For a circular orbit,  $r$  is constant, which means  $\dot{r} = 0$ ,  $\ddot{r} = 0$ . If the angular velocity is also constant, then  $\dot{\theta} = \omega$ ,  $\ddot{\theta} = 0$ . Therefore,  $\vec{a} = -r\omega^2\hat{r}$ . The minus sign indicates that the acceleration points toward the center.

## 1.6 Centripetal Force

For uniform circular motion, we just obtained  $\vec{a} = -r\omega^2\hat{r}$ . According to Newton's second law,  $\vec{F} = m\vec{a}$ . Therefore,  $\vec{F} = -mr\omega^2\hat{r}$ . If we consider only the magnitude of the force,

$$\vec{F}_c = mr\omega^2.$$

Recall from 1.2 that  $v = r\omega$ . Therefore,  $\omega = \frac{v}{r}$ . Substitute this into the centripetal-force equation

$$\vec{F}_c = \frac{mv^2}{r}.$$

Gravitational force is defined as  $F_g = G\frac{Mm}{r^2}$ , where  $M$  and  $m$  are the two masses,  $r$  is the distance between them, and  $G$  is the gravitational constant. For the satellite example, gravity provides the force required for circular motion. Therefore,  $F_c = F_g$ . Thus  $mr\omega^2 = G\frac{Mm}{r^2}$ . Canceling  $m$  from both sides gives  $r\omega^2 = G\frac{M}{r^2}$ . Multiplying by  $R^2$ , we obtain  $r^3\omega^2 = GM$ . Here, gravity acts as the centripetal force that keeps the satellite in circular motion.

## 2 The Three-Body Problem

### 2.1 Gravitational Force Between Two Bodies

Before considering three bodies, consider two masses  $m_1$  and  $m_2$  at positions  $r_1$  and  $r_2$ . The displacement vector pointing from body 1 to body 2 is  $\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$  and the distance between the two bodies is  $r_{12} = \|\vec{r}_{12}\| = \|\vec{r}_2 - \vec{r}_1\|$ .

For example, in two dimensions,  $\vec{r}_1 = (x_1, y_1)$ ,  $\vec{r}_2 = (x_2, y_2)$ . So,  $\vec{r}_{12} = (x_2 - x_1, y_2 - y_1)$  and  $r_{12} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ . The corresponding unit vector pointing from body 1 toward body 2 is  $\hat{r}_{12} = \frac{\vec{r}_{12}}{r_{12}} = \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|}$ .

According to Newton's law of universal gravitation, the magnitude of the gravitational force between the two bodies is  $\vec{F}_{12} = G\frac{m_1m_2}{r_{12}^2}\vec{r}_{12}$ . Since the force acting on body 1 points toward body 2, its vector form is  $\vec{F}_{12} = F_{12}\hat{r}_{12}$ . Substitute the two equations:  $\vec{F}_{12} = G\frac{m_1m_2}{r_{12}^2}\frac{\vec{r}_{12}}{r_{12}} = Gm_1m_2\frac{\vec{r}_{12}}{r_{12}^3}$ . Thus,  $\vec{F}_{12} = Gm_1m_2\frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3}$ .

### 2.2 From Gravitational Force to an Equation of Motion

According to Newton's second law,  $\vec{F} = m\vec{a}$ . Therefore, for body 1,  $m_1\vec{a}_1 = \vec{F}_{12}$ . Substituting the gravitational force gives  $m_1\vec{a}_1 = Gm_1m_2\frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3}$ . We have,  $\vec{a}_1 = Gm_2\frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3}$ . Notice something interesting:  $m_1$  disappears. The acceleration of body 1 due to body 2 depends on the mass  $m_2$ , the positions of the two bodies, and the gravitational constant  $G$ .

Since the positions  $r_1$  and  $r_2$  change with time, they can be written as  $r_1(t)$  and  $r_2(t)$ . Only their initial positions and velocities are known. The equation of motion determines how their positions evolve with time.

By definition, acceleration is  $\vec{a}_1 = \frac{d^2\vec{r}_1}{dt^2}$ . Therefore,  $\frac{d^2\vec{r}_1}{dt^2} = Gm_2\frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3}$ .

## 2.3 Initial Value Problem

We obtained

$$\frac{d^2 \vec{r}_1}{dt^2} = Gm_2 \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3}.$$

The unknown is not simply  $\vec{r}_1$ . It is the function  $\vec{r}_1(t)$  which describes the position of body 1 at every time  $t$ . Knowing the differential equation alone is not enough to determine a unique trajectory. Consider the much simpler equation  $\frac{d^2 x}{dt^2} = 0$ . Integrating once gives  $\frac{dx}{dt} = C_1$ , and integrating again gives  $x(t) = C_1 t + C_2$ . There are two unknown constants. To determine them, we need two pieces of information. In mechanics, these naturally correspond to  $x(0) = x_0$  and  $\dot{x} = v_0$ . Then  $x(t) = v_0 t + x_0$ . So for a second-order equation of motion, we normally specify both the initial position and initial velocity.

For body 1, we need

$$\vec{r}_1(0) = \vec{r}_{1,0}, \quad \dot{\vec{r}}_1(0) = \vec{v}_{1,0}.$$

Similarly for body 2, we need

$$\vec{r}_2(0) = \vec{r}_{2,0}, \quad \dot{\vec{r}}_2(0) = \vec{v}_{2,0}.$$

This combination of **differential equations + initial conditions** is called an Initial Value Problem (IVP).

## 2.4 Equations of Motion for Three Bodies

Consider three bodies with masses  $m_1, m_2, m_3$  and positions  $\vec{r}_1(t), \vec{r}_2(t), \vec{r}_3(t)$ . For body 1, there are now two gravitational forces acting on it:  $\vec{F}_{12}$  from body2 and  $\vec{F}_{13}$  from body3. By the principle of superposition,  $\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$ . Therefore,  $\vec{F}_1 = Gm_1 m_2 \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3} + Gm_1 m_3 \frac{\vec{r}_3 - \vec{r}_1}{\|\vec{r}_3 - \vec{r}_1\|^3}$ . Using Newton's second law, we obtain

$$\frac{d^2 \vec{r}_1}{dt^2} = Gm_2 \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3} + Gm_3 \frac{\vec{r}_3 - \vec{r}_1}{\|\vec{r}_3 - \vec{r}_1\|^3}.$$

Exactly the same reasoning gives

$$\frac{d^2 \vec{r}_2}{dt^2} = Gm_1 \frac{\vec{r}_1 - \vec{r}_2}{\|\vec{r}_1 - \vec{r}_2\|^3} + Gm_3 \frac{\vec{r}_3 - \vec{r}_2}{\|\vec{r}_3 - \vec{r}_2\|^3},$$

$$\frac{d^2 \vec{r}_3}{dt^2} = Gm_1 \frac{\vec{r}_1 - \vec{r}_3}{\|\vec{r}_1 - \vec{r}_3\|^3} + Gm_2 \frac{\vec{r}_2 - \vec{r}_3}{\|\vec{r}_2 - \vec{r}_3\|^3}.$$

Now we have the three-body equations of motion.

In two dimensions, each position vector contains two components,  $\vec{r}_i(t) = (x_i(t), y_i(t))$ , so the three vector equations represent six coupled second-order ODEs.

## 2.5 Converting Second-Order ODEs to First-Order ODEs

For each body,  $\vec{v}_i(t) = \frac{d\vec{r}_i}{dt}$ . And because acceleration is the derivative of velocity,  $\frac{d\vec{v}_i(t)}{dt} = \frac{d^2\vec{r}_i}{dt^2}$ . Therefore, the three second-order equations become six vector first-order equations:

$$\begin{aligned}\frac{d\vec{r}_1}{dt} &= \vec{v}_1, & \frac{d\vec{v}_1}{dt} &= Gm_2 \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3} + Gm_3 \frac{\vec{r}_3 - \vec{r}_1}{\|\vec{r}_3 - \vec{r}_1\|^3} \\ \frac{d\vec{r}_2}{dt} &= \vec{v}_2, & \frac{d\vec{v}_2}{dt} &= Gm_1 \frac{\vec{r}_1 - \vec{r}_2}{\|\vec{r}_1 - \vec{r}_2\|^3} + Gm_3 \frac{\vec{r}_3 - \vec{r}_2}{\|\vec{r}_3 - \vec{r}_2\|^3} \\ \frac{d\vec{r}_3}{dt} &= \vec{v}_3, & \frac{d\vec{v}_3}{dt} &= Gm_1 \frac{\vec{r}_1 - \vec{r}_3}{\|\vec{r}_1 - \vec{r}_3\|^3} + Gm_2 \frac{\vec{r}_2 - \vec{r}_3}{\|\vec{r}_2 - \vec{r}_3\|^3}\end{aligned}$$

In two dimensions, each position and velocity vector contains two components,

$$\vec{r}_i(t) = (x_i(t), y_i(t)) \quad \vec{v}_i(t) = (v_{xi}(t), v_{yi}(t)),$$

The distances between the three pairs of bodies are defined as  $r_{12}$ ,  $r_{13}$ , and  $r_{23}$ :

$$\begin{aligned}r_{12} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \\ r_{13} &= \sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2}, \\ r_{23} &= \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2}.\end{aligned}$$

Thus, the twelve first-order ODEs are

$$\begin{aligned}\dot{x}_1 &= v_{x1}, \\ \dot{y}_1 &= v_{y1}, \\ \dot{x}_2 &= v_{x2}, \\ \dot{y}_2 &= v_{y2}, \\ \dot{x}_3 &= v_{x3}, \\ \dot{y}_3 &= v_{y3}, \\ \dot{v}_{x1} &= Gm_2 \frac{x_2 - x_1}{r_{12}^3} + Gm_3 \frac{x_3 - x_1}{r_{13}^3}, \\ \dot{v}_{y1} &= Gm_2 \frac{y_2 - y_1}{r_{12}^3} + Gm_3 \frac{y_3 - y_1}{r_{13}^3}, \\ \dot{v}_{x2} &= Gm_1 \frac{x_1 - x_2}{r_{12}^3} + Gm_3 \frac{x_3 - x_2}{r_{23}^3}, \\ \dot{v}_{y2} &= Gm_1 \frac{y_1 - y_2}{r_{12}^3} + Gm_3 \frac{y_3 - y_2}{r_{23}^3}, \\ \dot{v}_{x3} &= Gm_1 \frac{x_1 - x_3}{r_{13}^3} + Gm_2 \frac{x_2 - x_3}{r_{23}^3}, \\ \dot{v}_{y3} &= Gm_1 \frac{y_1 - y_3}{r_{13}^3} + Gm_2 \frac{y_2 - y_3}{r_{23}^3}.\end{aligned}$$

## 2.6 Figure-Eight Orbit

This is a periodic symmetric orbit where the particles follow a figure eight trajectory in the plane. The orbit is such that the three masses chase one another around a single figure-eight shaped curve. The initial values for the periodic three-body figure-eight orbit, computed numerically by Carles Simó, place the three equal masses at specific positions and velocities.

System Constants:  $m_1 = m_2 = m_3 = 1$ , and  $G = 1$

Orbital Period:  $T \approx 6.32591398$

Initial Positions:

$$\begin{aligned}\vec{r}_1(0) &= (-0.97000436, 0.24308753), \\ \vec{r}_2(0) &= (0.97000436, -0.24308753), \\ \vec{r}_3(0) &= (0.0, 0.0).\end{aligned}$$

Initial Velocities:

$$\begin{aligned}\vec{v}_1(0) &= (0.466203685, 0.432365730), \\ \vec{v}_2(0) &= (0.466203685, 0.432365730), \\ \vec{v}_3(0) &= (-0.93240737, -0.86473146).\end{aligned}$$

## 2.7 Lagrange Points

The Lagrange points, also called the Lagrangian points or libration points, are points of equilibrium for small-mass objects under the gravitational influence of two massive orbiting bodies. At the Lagrange points, the gravitational forces of the two large bodies and the centrifugal pseudo-force balance each other. This makes Lagrange points excellent locations for satellites, as orbit corrections, and hence fuel requirements, needed to maintain the desired orbit are kept at a minimum.

The  $L1$  point lies on the line defined between the two large masses  $M1$  and  $M2$ . The  $L2$  point lies on the line through the two large masses beyond the smaller of the two. The  $L3$  point lies on the line defined by the two large masses, beyond the larger of the two. The  $L4$  and  $L5$  points lie at the third vertices of the two equilateral triangles in the plane of orbit whose common base is the line between the centers of the two masses.

Example: Sun and Earth.

$$M_{sun} = 1.989 \times 10^{30} \text{kg},$$

$$M_{earth} = 5.972 \times 10^{24} \text{kg},$$

$$D = 1.496 \times 10^8 \text{km}.$$

Suppose we initially place the Sun at  $x_{sun} = 0$ , and Earth at  $x_{earth} = D$ . The center of mass is

$$x_{cm} = \frac{M_{sun}x_{sun} + M_{earth}x_{earth}}{M_{sun} + M_{earth}}.$$

Since  $x_{sun} = 0$  and  $x_{earth} = D$ ,  $x_{cm} = \frac{M_{earth}D}{M_{sun} + M_{earth}} \approx 449 \text{km}$ . We now shift everything left by  $x_{cm}$  then  $x'_{sun} = -x_{cm} \approx -449 \text{km}$  and  $x'_{earth} = D - x_{cm} \approx 149599551 \text{km}$ . For normalization, we set the Sun–Earth distance to  $D = 1$  and define the dimensionless mass parameter

$$\mu = \frac{M_{earth}}{M_{sun} + M_{earth}} \approx 3.003 \times 10^{-6}.$$

Therefore, after shifting the origin to the center of mass and normalizing the Sun–Earth distance,  $x_{sun} = -\mu$  and  $x_{earth} = 1 - \mu$

Now, consider two bodies orbiting their common center of mass. Suppose we have two masses  $M_1$  and  $M_2$ , separated by distance  $R$ . Their distances from the center of mass are  $r_1$  and  $r_2$ , so  $r_1 + r_2 = R$ . From the definition of center of mass,  $M_1 r_1 = M_2 r_2$ . Solving these gives

$$r_1 = \frac{M_2}{M_1 + M_2} R \quad \text{and} \quad r_2 = \frac{M_1}{M_1 + M_2} R.$$

The gravitational force pulling  $M_2$  toward  $M_1$  is  $F_g = \frac{GM_1 M_2}{R^2}$ . The centripetal force is  $F_c = M_2 r_2 \omega^2$ . Therefore,  $\frac{GM_1 M_2}{R^2} = M_2 r_2 \omega^2$ , which simplifies to

$$\frac{GM_1}{R^2} = r_2 \omega^2.$$

Substituting  $r_2 = \frac{M_1}{M_1 + M_2} R$ , we obtain  $\frac{GM_1}{R^2} = \frac{M_1}{M_1 + M_2} R \omega^2$ . Multiply both sides by  $(M_1 + M_2)$ , we obtain  $\frac{G(M_1 + M_2)}{R^2} = R \omega^2$ . Finally, we get

$$\omega^2 = \frac{G(M_1 + M_2)}{R^3}.$$

For normalization, we choose the units such that  $R = D = 1$ ,  $\omega = 1$ , and  $M_1 + M_2 = 1$ . Substituting these normalized values into the previous equation gives  $1^2 = \frac{G(1)}{1^3}$ . Thus,  $1 = G$ . Now, we have

$$G = 1, \quad \omega = 1, \quad M_1 = (1 - \mu), \quad M_2 = \mu, \\ (x_{sun}, y_{sun}) = (-\mu, 0), \quad (x_{earth}, y_{earth}) = (1 - \mu, 0)$$

with  $\mu \approx 3.003 \times 10^{-6}$ .

### L1 position:

We place the satellite at  $(x, 0)$ . The distance from the satellite to the Sun is  $x + \mu$ . The distance from the satellite to the Earth is  $(1 - \mu) - x$ . Now consider the three terms.

1. **Sun pulls the satellite to the left**, so its gravitational acceleration is  $-\frac{(1-\mu)}{(x+\mu)^2}$ .
2. **Earth pulls the satellite to the right**, so its gravitational acceleration is  $\frac{\mu}{(1-\mu-x)^2}$ .
3. **The centrifugal acceleration** is  $\omega^2 r = x$ .

Therefore, at  $L1$

$$x - \frac{(1-\mu)}{(x+\mu)^2} + \frac{\mu}{(1-\mu-x)^2} = 0.$$

Notice: Since the satellite mass appears in every force term, it cancels when applying Newton's second law. Therefore, we work directly with acceleration.

Newton's Method converges to approximately  $x \approx 0.990027$ . Earth is at  $1 - \mu \approx 0.999997$ . So the normalized distance between Earth and  $L1$  is  $\approx 0.009970$ . Now convert back to kilometers  $D = 1.496 \times 10^8$  km. Therefore,  $d_{L1} = 0.009970 \times D \approx 1.49 \times 10^6$  km. So  $L1$  is about 1.49 million km from Earth toward the Sun.

**L2 position:**

We place the satellite at  $(x, 0)$ . The distance from the satellite to the Sun is  $x + \mu$ . The distance from the satellite to the Earth is  $x - (1 - \mu)$ . Now consider the three terms.

1. **Sun pulls the satellite to the left**, so its gravitational acceleration is  $-\frac{(1-\mu)}{(x+\mu)^2}$ .
2. **Earth pulls the satellite to the left**, so its gravitational acceleration is  $-\frac{\mu}{(x-1+\mu)^2}$ .
3. **The centrifugal acceleration** is  $\omega^2 r = x$ .

Therefore, at  $L2$

$$x - \frac{(1-\mu)}{(x+\mu)^2} - \frac{\mu}{(x-1+\mu)^2} = 0.$$

Newton's Method converges to approximately  $x \approx 1.010034$ . Earth is at  $1 - \mu \approx 0.999997$ . So the normalized distance between Earth and  $L2$  is  $\approx 0.010037$ . Now convert back to kilometers  $D = 1.496 \times 10^8$  km. Therefore,  $d_{L2} = 0.010037 \times D \approx 1.50 \times 10^6$  km. So  $L2$  is about 1.5 million km from Earth toward the Sun.

**L3 position:**

We place the satellite at  $(x, 0)$ . The distance from the satellite to the Sun is  $-\mu - x$ . The distance from the satellite to the Earth is  $(1 - \mu) - x$ . Now consider the three terms.

1. **Sun pulls the satellite to the right**, so its gravitational acceleration is  $\frac{(1-\mu)}{(-\mu-x)^2}$ .
2. **Earth pulls the satellite to the right**, so its gravitational acceleration is  $\frac{\mu}{(1-\mu-x)^2}$ .
3. **The centrifugal acceleration** is  $\omega^2 r = x$ .

Therefore, at  $L3$

$$x + \frac{(1-\mu)}{(-\mu-x)^2} + \frac{\mu}{(1-\mu-x)^2} = 0.$$

Newton's Method converges to approximately  $x \approx -1.00000125$ . Sun is at  $-\mu$ . So the normalized distance between Sun and  $L3$  is  $\approx 0.99999825$ . Now convert back to kilometers  $D = 1.496 \times 10^8$  km. Therefore,  $d_{L3} = 0.99999825 \times D \approx 1.496 \times 10^8$  km. So  $L3$  is about 149.6 million km from the Sun on the side opposite Earth.

**L4, L5 positions:**

The Sun, Earth, and  $L4$  or  $L5$  form an equilateral triangle. Since we normalized the Sun–Earth distance,  $D = 1$ . Therefore, all three sides of each equilateral triangle have length 1.

The x-coordinate of  $L4$  and  $L5$  is exactly halfway between the Sun and Earth,  $x = \frac{-\mu+(1-\mu)}{2} = \frac{1}{2} - \mu$ . For our Sun–Earth system,  $x \approx 0.499997$ .

To find the y-coordinate, the equilateral triangle is divided into two right triangles. Using the Pythagorean theorem,  $y^2 + (\frac{1}{2})^2 = 1$ . Thus,  $y = \pm \frac{\sqrt{3}}{2}$ .

Converting the normalized coordinates back to kilometers by multiplying by  $D \approx 1.496 \times 10^8$  km, we obtain  $L4 = (7.480 \times 10^7, 1.296 \times 10^8)$  km and  $L5 = (7.480 \times 10^7, -1.296 \times 10^8)$  km.

**Code explanation:**

We place a satellite at  $L1$  in the Sun-Earth system. First, we calculate the dimensionless mass parameter  $\mu$  and the angular velocity  $\omega$  of the Sun-Earth system. We have

$$G = 6.67430 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2},$$

$$D = 1.495978707 \times 10^{11} \text{ m},$$

$$M1 = 1.98847 \times 10^{30} \text{ kg}, \quad M2 = 5.9722 \times 10^{24} \text{ kg}$$

thus,

$$\mu = \frac{M_2}{M_2 + M_1}, \quad \omega = \sqrt{\frac{G(M_1 + M_2)}{D^3}}.$$

The distances from  $x$  to the Sun and Earth are

$$r_1 = |x + \mu|, \quad r_2 = |x - (1 - \mu)|.$$

The general equilibrium equation of  $L1$ ,  $L2$ , and  $L3$  is

$$x - (1 - \mu) \frac{x + \mu}{|\mu + x|^3} - \mu \frac{x - (1 - \mu)}{|x - (1 - \mu)|^3} = 0.$$

This equation is implemented in **collinear\_equation(x)**.

Since the equilibrium equation is nonlinear, we use **root\_scalar()** to numerically determine the positions of  $L1$ ,  $L2$ , and  $L3$ . Recall that in normalized coordinates,  $x_{sun} = -\mu$ , and  $x_{earth} = 1 - \mu$ . Therefore, the three regions are

$$L1 : -\mu < x < 1 - \mu,$$

$$L2 : x > 1 - \mu,$$

$$L3 : x < -\mu.$$

$L4$  and  $L5$  are much easier, their coordinates are simply  $(\frac{1}{2} - \mu, \pm \frac{\sqrt{3}}{2})$ .

We set the initial positions and velocities of the Sun, Earth, and satellite.

$$(x_1, y_1) = (-D\mu, 0.0), \quad (x_2, y_2) = (D(1 - \mu), 0.0), \quad (x_3, y_3) = (D(x_{L1}), 0.0),$$

$$(v_{x1}, v_{y1}) = (0.0, \omega x_1), \quad (v_{x2}, v_{y2}) = (0.0, \omega x_2), \quad (v_{x3}, v_{y3}) = (0.0, \omega x_3).$$

The motion of the Sun, Earth, and satellite is described by a system of twelve first-order ODEs. The system is simulated for one year. Since the ODE system uses SI units, the time interval is converted from days to seconds. 1 day = 24.0\*60.0\*60.0 s. Therefore,  $t_{start} = 0$ , and  $t_{end} = 365.25*1$  day.

**np.linspace()** generates 1500 equally spaced time points between  $t_{start}$  and  $t_{end}$ . The numerical solution will be stored at these time points and later used for plotting and animation.

The twelve first-order ODEs are solved numerically using **solve\_ivp()**, whose default method is RK45.